

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

EEE 221 — Past Questions & Materials
by Renance

Course: EEE 221
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 6
Levels: 200

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Paper 1 of 6 — EEE 221 BASIC ELECTRICAL ENGINEERING II 2023I2024

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

EXAMINATION: Second Semester
 SESSION: 2023/2024
 DATE: 24/09/2024



COURSE CODE: EEE 221
 COURSE TITLE: Basic Electrical Engineering II
 TIME ALLOWED: 2 ½ Hrs.

INSTRUCTIONS:

- I. Questions 1 and 2 in SECTION A are **COMPULSORY**
- II. Answer **ONLY ONE** question from SECTION B
- III. Answer **ONLY ONE** question from SECTION C
- IV. Answer **FOUR** questions in all

SECTION A

1. (a) A 415 V, 3-phase, 4-wire star connected system supplies 3 loads as follows; Red-phase – 8kW, Yellow-phase – 1.2kW and Blue-phase – 0.8kW. Represent the system with a diagram and determine the following:
 - i. current in each line
 - ii. current in the neutral conductor.
- (b) Explain how a three-phase supply is generated
2. (a) The pressure of crude oil flow is monitored in a control room at a flow station. State the measurement technique involved in the process. (3 marks)
- (b) Briefly describe the analog instrument used in power measurement. (6 ½ marks)
- (c) Suppose a measurement was performed on a resistor and the following results were obtained.
 True Value = 5.25kΩ
 Measured Value = 5.13kΩ
 Calculate the relative error that occurs from the measurement. (8 marks)

SECTION B

3. (a) Discuss the working principles of a transformer and states the laws associated with it.
- (b) Explain with the aid of a diagram the term 'Equivalent Circuit of a Transformer'. Simplify the circuit and derive the formula for the induced Voltage E₂.
- (c) A single-phase transformer has 400 primary and 1000 secondary turns. The net cross-sectional area of the core is 60 cm². If the primary winding is connected to a 50Hz supply at 520V. Calculate the peak value of flux density in the core and the voltage induced in the secondary winding.
4. (a) Discuss the operations of a DC motor and state the laws associated with the rotation of a DC electric motor.
- (b) List and discuss the significance of Back EMF.
- (c) Discuss the difference between a lap and wave winding generator and show diagrams for both.
- (d) A 250V, 4 pole, wave wound DC series motor has 782 conductors on its armature. It has armature and series field resistance of 0.75 ohms. The motor takes a current of 40A. Estimate its speed and gross torque developed if it has a flux per pole of 25mWb.

SECTION C

5. (a) Using simple circuits describe how to forward bias and reverse bias a diode. (4 marks)
- (b) The single-phase halfwave rectifier of Fig. Q5(b) is supplied from a 220 V, 50 Hz source through the step-down transformer with turns ratio $n = 15:1$. The load resistance R_L is 5Ω. Determine:

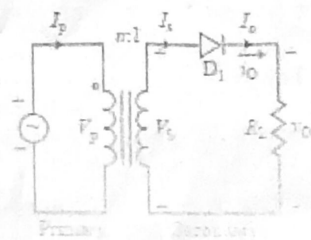


Figure Q5(b)

(6 ½ marks)

- (i) the average output voltage $V_o(av)$,
 (ii) the average load current $I_o(av)$,
 (iii) the rms load voltage $V_o(rms)$,
 (iv) the rms load current $I_o(rms)$,
 (v) the peak inverse voltage PIV of the diode,
 (vi) the DC output power $P_o(dc)$.

- (c) The circuit in Fig Q5(c) has the following parameters: $V_S = 12\text{ V}$; $V_B = 11\text{ V}$; $R_1 = 5\ \Omega$; $R_2 = 10\ \Omega$; $R_3 = 10\ \Omega$. Determine if the ideal diode is conducting. (7 marks)

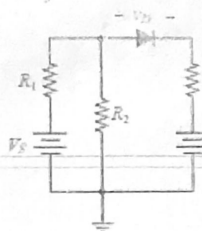


Figure Q5(c)

$V_1 = 12\text{ V}$
 $V_2 = 11\text{ V}$
 $R_1 = 5\ \Omega$
 $R_2 = 10\ \Omega$
 $R_3 = 10\ \Omega$

- 6) (a) For the Zener diode voltage regulator circuit shown in Figure Q6a, derive the expression for the maximum Zener current in terms of voltage and current only. (4 marks)

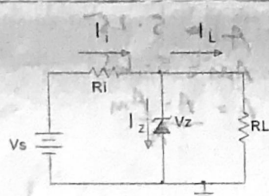


Figure Q6a

- (b) Design a voltage regulator circuit to power a car radio at $V_L = 8\text{ V}$ from an automobile battery whose voltage may vary between 11 and 13.6 V. The current in the radio will vary between 0 (off) to 100 mA (full volume). The minimum Zener current is one-tenth the maximum Zener current. (6 ½ marks)

- (c) For the circuit in Fig Q6c find the following:

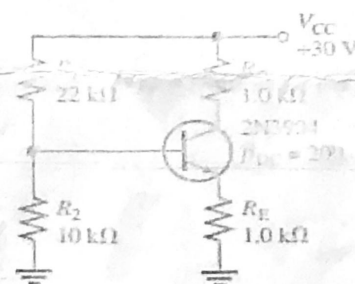


Figure Q6c

- i. Base voltage, V_B
 ii. Emitter voltage, V_E
 iii. emitter current, I_E
 iv. Collector current, I_C
 v. Base current, I_B
 vi. Collector voltage, V_C
 vii. Collector-Emitter voltage, V_{CE}

(7 marks)

Paper 2 of 6 — EEE 221 BASIC TECHNOLOGY ENGINEERING II

Level 200 • Second Semester • PastQuestion

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date

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF TECHNOLOGY

EXAMINATION: First Semester
COURSE CODE: EEE 211
TIME ALLOWED: 2 Hrs



SESSION: 2018/2019
COURSE TITLE: Basic Electrical Engineering I
DEGREE: B.ENG

INSTRUCTION: Answer FOUR Questions from both sections

SECTION A: Answer TWO Questions from this section

1. (a) State Maximum Power Transfer theorem, and justify the need for such a theorem. (4½ marks)
- (b) (i) State Superposition theorem. Explain why this theorem does not apply to power as an electrical quantity, and state how power can be computed using this theorem. (5 marks)
- (ii) Using Superposition theorem, calculate the current flowing through the $4\ \Omega$ resistor in figure Q1 (b). Also, compute the voltage drop and power dissipated in the $4\ \Omega$ resistor. (8 marks)

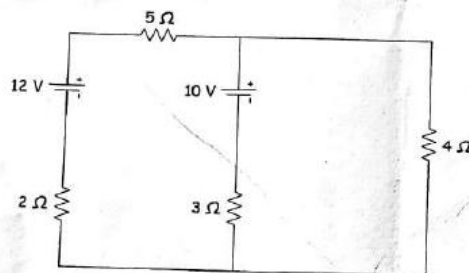


Figure Q1 (b)

2. (a) Explain the following terms: (i) Node (ii) Loop (iii) Branch (iv) Transient response. (6 marks)
- (b) For the circuit of figure Q2 (b), (i) Write out the differential equation (ii) Compute the time constant, natural response and forced response (3½ marks)
- (c) Calculate the mesh currents and branch currents in the circuit of figure Q2 (c). (8 marks)

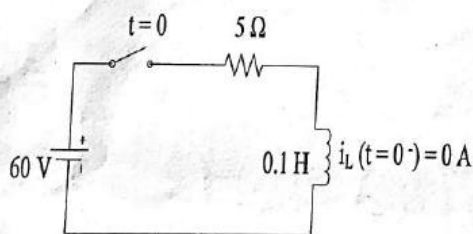


Figure Q2 (b)

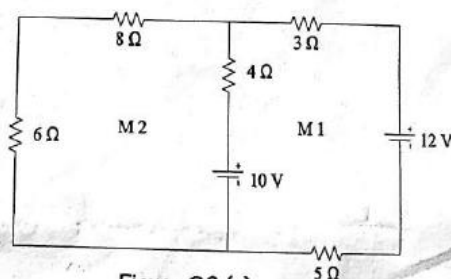


Figure Q2 (c)

3. (a) State Kirchhoff's current law and Kirchhoff's voltage law. (3 marks)
- (b) State Norton's theorem. (2½ marks)
- (c) (i) Using Thevenin theorem, calculate the current through the $8\ \Omega$ resistor in figure Q3 (c). (9 marks)
- (ii) Draw the Norton equivalent circuit of the circuit obtained in Question 3c (i) and verify the value of the current flowing through the $8\ \Omega$ resistor. (3 marks)

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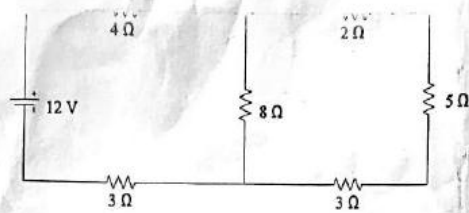


Figure Q3 (c)

SECTION B: Answer TWO Questions from this section

4. (a) State the laws of magnetism. (3½ marks)
- (b) Define the following: (i) magnetic field (ii) magnetic flux (iii) relative permeability (6 marks)
- (c) An inductive coil of resistance $10\ \Omega$ and inductance 0.1 H is connected in parallel with a $150\ \mu\text{F}$ capacitor to a variable frequency, 200 V supply. Find the resonant frequency at which the total current taken from the supply is in phase with the supply voltage. Find also the value of this current. Draw the circuit arrangement and the phasor diagrams. (8 marks)
5. (a) With the aid of a circuit diagram and all necessary equations, show that when an alternating voltage is applied to a purely capacitive circuit, the resulting current leads the applied voltage by $\frac{\pi}{2}$ radians. (8 marks)
- (b) A magnetic circuit is excited by three coils as shown in figure Q5 (b). Calculate the flux produced in the air gap. The material used for core is iron having relative permeability of 800. The length of the magnetic circuit is 100 cm with an air gap of 2 mm in it. The core has uniform cross-section of 6 cm^2 . Take $\mu_0 = 4\pi \times 10^{-7}$. (9½ marks)

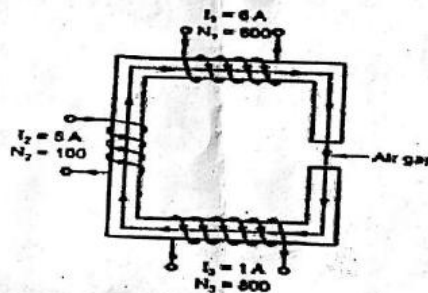


Figure Q5 (b)

6. (a) Enumerate the importance of RMS value in the field of Engineering. (6 marks)
- (b) The waveform of the voltage and current of a circuit are given by:
 $e = 120\sin(314t)$ and $i = 10\sin(314t + \frac{\pi}{6})$. Calculate the values of the resistance and capacitance which are connected in series to form the circuit. Also draw waveforms for current, voltage and phasor diagram. Calculate power consumed by the circuit. (11½ marks)

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Paper 3 of 6 — EEE 221 BASIC ELECTRICAL ENGINEERING 11 2017,2018

Level 200 • Second Semester • PastQuestion

| _ FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

' COLLEGE OF TECHNOLOGY

—EXAMINATION: 2nd Semester COURSE CODE: EEE 221,

SESSION: 2017/2018

COURSE TITLE: Basic Electrical Engineering I!

DEGREE: B. Eng. ; : : ~ TIME ALLOWED; 2 1/2 Hours :

a Ac

INSTRUCTION: Answer any four (4) questions. (er ia

rr we

ae is the study of semiconductor materials important in electronic engineering? ¥

Ab) Explain with suitable diagram how to properly bias a transistor ; :

(c) (i) With the aid of a labeled diagram, describe the basic components of an AC generator connected to an external circuit.

(ii) State the difference between DC and-AC generator,

(ili) Differentiate single phase and three phase voltages with respect to how they are generated.

(25 Marks)

in the following;

(i) Hole theory (ii) Doping and its importance (iii) Intrinsic material

Using appropriate diagram explain how

(i) An N-type semiconductor is produced (ii) A P-type semiconductor is produced

(c) A single phase transformer has 250 turns in the primary and 600 turns in the secondary. The cross-sectional area of

the core is 40cm². If the primary winding is connected to a 50Hz supply to 250V. Calculate: (i) Peak flux density.

(ii) Voltage induced in the secondary winding. (25 Marks)

define the term "semiconductor", ; f

b) Draw and explain any two (2) configuration of a transistor as an amplifier and the form of amplification that will take

place in each configuration. :

(c) (i) What is a transformer? : SA SE, emf talS eS

~ (ii) Derive the emf equation of a transformer with respect to the rms value of emf induced in the primary and secondary

winding. (25 Marks) ~

4, (a) Using the atomic structure of silicon and Germanium why they are classified as semiconductor material

—.. (b) Draw and explain the characteristic curve of a Zener Diode : :

Using the AC signal waveform shown below, knowing the Y-gain was set to 10 per division, Calculate:

Cee

Pe=Xep Pref

Pm Sey beSang ~

(i) The peak to peak voltage (ii) The peak voltage (V_p) (iii) The Peak current (I_p) using a resistance of 3Ω (iv) The root

mean square values of p.d (V_{rms}) and current (I_{rms}) (28 Marks)

Period (T)

ing appropriate diagram, explain briefly

(a) How a diode is produced (b) Forward and Reverse biasing of a diode — (c) Applications of a diode

(d) Characteristic curve of a silicon diode

— (e) A balanced star connected load is supplied from a symmetrical 3-phase, 205V system: The current in each phase is 15A

15° behind the phase voltage. Find: (i) Phase voltage (ii) Total power and (iii) Reactive power drawn by load.

(a) What is a Zener Voltage :

(b) What is the difference between a Zener Diode and the Ordinary Diode

(c) How does temperature affect the characteristics of a semiconductor material? nie

(i) How do you connect an ammeter to measure electric current through a wire or line of a circuit?

(ii) Draw, label and describe the wattmeter in terms of the two coils with respect to how they are connected.

(i) List four methods of measuring inductance of an inductor. (25 marks) Ep OP die

Paper 4 of 6 — EEE 221 PAST QUESTION

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

EXAMINATION: Second Semester
SESSION: 2020/2021
DATE: November 26, 2022



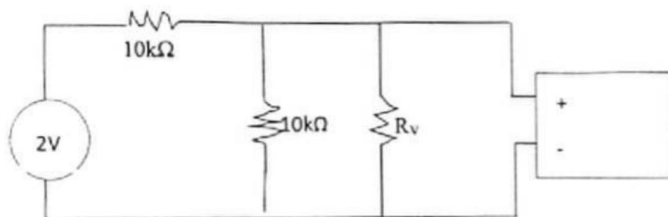
COURSE CODE: EEE 221
COURSE TITLE: Basic Electrical Engineering II
TIME ALLOWED: 2 ¼ Hrs.

INSTRUCTION: Answer four questions in all. Answer at least one question from each section

SECTION A

QUESTION ONE

- (a) An electromechanical Voltmeter with an internal resistance of $3k\Omega$ and an electronic voltmeter with an internal resistance of $30M\Omega$ are used separately to measure potential differences between point A and the ground of the circuit below

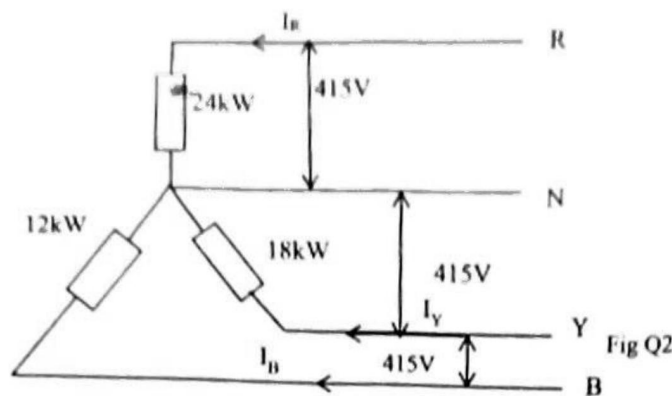


Calculate the voltages that will be indicated by each of the two instruments and the percentage error in each case.

- (b) (i) What are instrument loading effects?
(ii) List 4 methods of measuring the inductance of an inductor.

QUESTION TWO

- (a) What are the advantages of three-phase system?
(b) Figure Q2 shows a 415V, 3 phases 4 wire star-connected system supplying purely resistive loads. Determine the following:
i. Current in each line
ii. Current in the neutral conductor.



SECTION B

QUESTION 3

- Discuss briefly the constructional features of a DC machine.
- Show that the emf equation of a DC machine is given by $E = \frac{\Phi Z N}{60} \times \frac{P}{A}$ Volts.
- A 120kVA, 11kV/ 440V, 50Hz single phase transformer has an effective cross-sectional area of 0.04 m^2 and has 200 turns in its low voltage (LV) winding. Determine:
 - The maximum value of flux density
 - The number of turns on the high voltage (HV) winding and
 - Full load current in each winding.

QUESTION 4

- Enumerate and discuss briefly, the two types of transformer construction known to you.
- Briefly discuss the following types of single phase induction motors:
 - Capacitor start motor
 - Capacitor start and run motor
- A 3-phase, 16pole, alternator has an air-gap flux of 0.05Wb per pole, simultaneously distributed. The stator has 2 slots per pole per phase and 4 conductors per slot. The coil span is 150° . Calculate the phase and line-induced emf when it is driven at 375rpm.

SECTION C

QUESTION 5

- Using simple circuits describe how to forward bias and reverse bias a diode. (4 marks)

(b) The single-phase half-wave rectifier of Fig. Q5(b) is supplied from a 220 V, 50 Hz source through the step-down transformer with a turns ratio $n = 15:1$. The load resistance R_L is 5Ω . Determine:

- the average output voltage $V_o(av)$,
 - the average load current $I_o(av)$,
 - the rms load voltage $V_o(rms)$,
 - the rms load current $I_o(rms)$,
 - the peak inverse voltage PIV of the diode,
 - the DC output power $P_o(dc)$.
- (6 ½ marks)

(c) The circuit in Fig Q5(c) has the following parameters: $V_S = 12\text{ V}$; $V_B = 11\text{ V}$; $R_1 = 5\Omega$; $R_2 = 10\Omega$; $R_3 = 10\Omega$. Determine if the ideal diode is conducting. (7 marks)

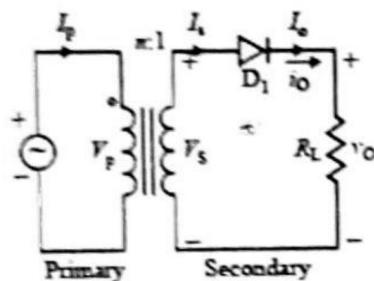


Figure Q5(b)

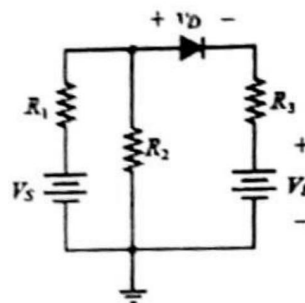


Figure Q5(c)

QUESTION 6

(a) For the Zener diode voltage regulator circuit shown in Figure Q6a, derive the expression for the maximum Zener current in terms of voltage and current only. (4 marks)

(b) Design a voltage regulator circuit to power a car radio at $V_L = 8\text{ V}$ from an automobile battery whose voltage may vary between 11 and 13.6 V. the current in the radio will vary between 0 (off) to 100 mA (full volume). The minimum Zener current is one-tenth the maximum Zener current. (6 ½ marks)

(c) For the circuit in Fig Q6c find the following:

- Base voltage, V_B
- Emitter voltage, V_E
- Emitter current, I_E
- Collector current, I_C

- v. Base current, I_B
- vi. Collector voltage, V_C
- vii. Collector-Emitter voltage, V_{CE}

(7 marks)

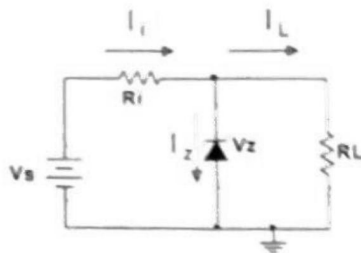


Figure Q6b

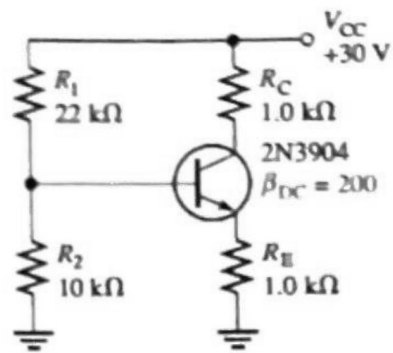


Figure Q6c

Paper 5 of 6 — EEE 221

Level 200 • Second Semester • Handout

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN

COLLEGE OF ENGINEERING AND TECHNOLOGY

SECOND SEMESTER

DEPARTMENT: PETROLEUM ENGG ACADEMIC SESSION: 2023/2024

COURSE TITLE: BASIC ELECTRICAL ENGINEERING II COURSE CODE: EEE221

COURSE CREDIT: 2 LEVEL: 200

S/N NAMES MAT NO CA EXAM TOTAL GRADE SUMMARY

1 ADAMS DIVINE TARAMA-EBI 10022947JC 1 1 2 F GRADE NUMBERS

2 AFENFIA OYINNUA EZEKIEL COET/9475/2022 15 27 42 E A 25

3 AGBOR SAMUEL JOHN COET/8595/2021 21 19 40 E B 50

4 AGHWANA SAMUEL AYOMIDE COET/10303/2022 16 24 40 E C 47

5 AJIFE GEORGE EJIROGHENE COET/9707/2022 20 21 41 E D 21

6 ANIAGOLU CHIEMELIE VICTOR COET/8305/2021 26 41 67 B E 48

7 ANIMAM GIDEON ONYELONI COET/9939/2022 26 37 63 B F 13

8 ANTHONY ISRAEL ONORIODE COET/8635/2021 24 44 68 B TOTAL 204

9 ANYAECHE CHINAZA BLESSING COET/9553/2022 25 40 65 B

10 APPIAH VANESSA MERCY COET/9447/2022 23 38 61 B PASSED 191

11 AZIBAODI PETER ELLIOT-EBUTU COET/8331/2021 18 23 41 E FAILED 13

12 BAMIDELE STEPHEN OMOTAYO COET/8210/2021 19 27 46 D

13 BAMISILE ABDULQUDUS AYOMIDE COET/10244/2022 19 26 45 D %PASSED 94%

14 BINGILA FAVOUR COET/9643/2022 30 56 86 A %FAILED 6%

15 BONFERE EMMANUEL OSEIWE COET/9742/2022. 23 41 64 B

16 CHIMA OLISAEMEKA GERALD COET/9489/2022 17 23 40 E

17 CHINWEOKWU CHINENYE FAVOUR COET/8335/2021 22 46 68 B LECTURER: Engr. Dr. Akinloye B. O.

18 DASE OGHENEVWEDE CLAIRE COET/9619/2022 15 27 42 E

19 EKELE WISDOM ATTAH COET/9006 /2021 24 44 68 B

20 EKERUKE IMO BONIFACE COET/8686/2021 28 52 80 A LECTURER: Engr. Dr. Enalume K. O.

21 ELENBESUNU EVANGELYN UDEBAOSE COET/8470/2021 27 47 74 A

22 EMERUEM JOSEPH EBUKA COET/9426/2022 21 30 51 C

23 GODWIN FAVOUR EBUKA COET/10353/2022 20 31 51 C LECTURER: Engr. Dr. Efenedo

24 GODWIN TAMARADENYEFA GODSPOWER COET/9681/2022 18 24 42 E

25 IKEH DANIEL IKENNA COET/8564/2021 26 42 68 B

26 ILUFOYE TOYEEB ABIODUN COET/10201/2022 17 23 40 E LECTURER: Engr. Avworo

27 IMORAME ELIZABETH EHIZEFUA COET/9515/2022 24 37 61 B

28 IYOGHO OGHENEGARHE VICTOR COET/10149/2022 24 44 68 B

29 KILANI ABDUSSALAM ADEBARE COET/8964/2021 18 22 40 E H.O.D.: Engr. Dr. Enalume K. O.

30 MATTHEW EMMANUEL AZIBATOR COET/8236/2021 21 26 47 D

Page 1

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN
COLLEGE OF ENGINEERING AND TECHNOLOGY
SECOND SEMESTER

DEPARTMENT: PETROLEUM ENGG ACADEMIC SESSION: 2023/2024

COURSE TITLE: BASIC ELECTRICAL ENGINEERING II COURSE CODE: EEE221

COURSE CREDIT: 2 LEVEL: 200

S/N NAMES MAT NO CA EXAM TOTAL GRADE SUMMARY

31 MAXWELL-TENUMAH ARATUTUN COET/10417/2022 27 48 75 A GRADE NUMBERS

32 NGBAKOR VICTOR COET/8817/2021 25 40 65 B A 25

33 NNODO KENECHUKWU HENRY COET/10395/2022 21 19 40 E B 50

34 NSIANYA ECHEZONACHUKWU GAIUS COET/10102/2022 24 44 68 B C 47

35 Nwigwe Precious Somtochukwu COET/9626/2022 21 29 50 C D 21

36 OAIKHENA MICHELLE ZEMOYA COET/10205/2022 28 47 75 A E 48

37 OCHAI SOLOMON UNCHE COET/9816/2022 29 54 83 A F 13

38 OCHEI JOSEPH TEMIREMI COET/8827/2021 5 4 9 F TOTAL 204

39 ODILI CHUKWUKA FRANKLIN COET/9456/2022 27 49 76 A

40 ODO MAGNUS BOBUA COET/8962/2021 22 33 55 C PASSED 191

41 ODUMUYE OLUWADAMILARE DANIEL COET/9204/2021 17 24 41 E FAILED 13

42 OFFUNG CHARLES OSAGIE COET/8218/2021 27 48 75 A

43 OGAR FAVOUR BERNARD COET/10365/2022 21 32 53 C %PASSED 94%

44 OGBONNA VIVIAN ONYINYECHUKWU COET/9680/2022 29 52 81 A %FAILED 6%

45 OGENE IFUNANYA BLESSING COET/10454/2022 9 14 23 F

46 OGHENEKEVWE HOLY ONORIODE COET/9600/2022 18 30 48 D

47 OGHENEMIDERHI LENNON SPINNSON COET/10292/2022 20 20 40 E LECTURER: Engr. Dr. Akinloye B. O.

48 OKABA DESTINY AKWAJI COET/8748/2021 24 41 65 B

49 OKAFOR CHUKWUDI DAVID COET/10265/2022 20 38 58 C

50 OKAFOR DAVID EBUKA COET/9181/2021 21 46 67 B LECTURER: Engr. Dr. Enalume K. O.

51 OKORIE CHIDUBEM EMMANUEL COET/8439/2021 29 52 81 A

52 OKPE ERNEST EJIKE COET/7346/2020 15 27 42 E

53 OKWOLI BENJAMIN SUNDAY COET/8744/2021 21 33 54 C LECTURER: Engr. Dr. Efenedo

54 OLASEHINDE SAMUEL OLAJOY COET/8369/2021 20 21 41 E

55 OLATUNJI JOSEPH AYOMIDE COET/10228/2022 27 37 64 B

56 OMELIME OFURE TRUST COET/9459/2022 29 46 75 A LECTURER: Engr. Avworu

57 OMINIRAN DAVID OLUWANIFEMI COET/9604/2022 23 37 60 B

58 ONOJAH GODWIN UGBEDJO COET/8329/2021 22 39 61 B

59 ONUEGBU OSINAKACHI MISHAEL COET/8919/2021 27 50 77 A H.O.D.: Engr. Dr. Enalume K. O.

60 OPUTE LUCAS CHUKWUKADIBIE COET/9682/2022 26 41 67 B

Page 2

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN

COLLEGE OF ENGINEERING AND TECHNOLOGY

SECOND SEMESTER

DEPARTMENT: PETROLEUM ENGG ACADEMIC SESSION: 2023/2024

COURSE TITLE: BASIC ELECTRICAL ENGINEERING II COURSE CODE: EEE221

COURSE CREDIT: 2 LEVEL: 200

S/N NAMES MAT NO CA EXAM TOTAL GRADE SUMMARY

61 OSARENREN FRANCIS OSADEBAWEN COET/8941/2021 21 37 58 C GRADE NUMBERS

62 OSAZEE EUPHRASIA NORENSE COET/8589/2021 23 40 63 B A 25

63 OSSAI CHUKWUEMEKE ISMAIL COET/10275/2022 16 24 40 E B 50

64 OTOWOR ISAIAH COET/8320/2021 20 20 40 E C 47

65 OTUHOHI NAGEL ABOGHENE COET/10086/2022 12 15 27 F D 21

66 OTURUHOYI IRUOROBOGHENE GRACE COET/10356/2022 18 23 41 E E 48

67 OWENA OGHENESUE DAVID COET/9042/2021 18 27 45 D F 13

68 ROSELYN SAKANWI NJIRAAN COET/8339/2021 16 25 41 E TOTAL 204

69 SALAWUDEEN RUFAL COET/9448/2022 17 25 42 E

70 SHOFOLABO OLUWABAMISE CHARLES COET/8448/2021 21 34 55 C PASSED 191

71 SINCLAIR EBITIMI JOSHUA COET/9499/2022 19 22 41 E FAILED 13

72 STEPHEN OLUWADAMILOLA GIFT COET/9601/2022 30 52 82 A

73 SUCCESS OSAGIE OSAMAGBE COET/8414/2021 22 34 56 C %PASSED 94%

74 UBEKU AFORKE COET/10017/2022 24 32 56 C %FAILED 6%

75 UMADO MELODY OGHENEKEVWE COET/8231/2021 23 37 60 B

76 UMEH CLAIRE CHIMDUMEME COET/9455/2022 12 14 26 F

77 WARRI GODSPOWER COET/10426/2022 23 17 40 E LECTURER: Engr. Dr. Akinloye B. O.

78 WILSON SHALOM OGHENEKEVWE COET/9795/2022 17 28 45 D

79 YAKOVU EMMANUEL CHUKWUYENUM COET/8255/2021 23 37 60 B

80 OKUNDAYE ISABELLA OSARUGUE COET/11610/2023 17 29 46 D LECTURER: Engr. Dr. Enalume K. O.

81 IYAGI AMBROSE COET/11587/2023 19 24 43 E

82 ASANBIARA ISAIAH ADEBOYE COET/11569/2023 21 33 54 C

83 AGUNU DANIEL OGHENAOGIE COS/8884/2021 20 20 40 E LECTURER: Engr. Dr. Efenedo

84 EDUZOR CHUKWUNONSO COET/10466/2022 18 32 50 C

85 OLADIMEJI IBRAHEEM FUNSO COET/9636/2022 16 24 40 E

86 NWUTOBO GABRIEL NDUBUISI COET/10311/2022 22 46 68 B LECTURER: Engr. Avworu

87 ABAYE AYEBAEKIPRETEI ESEOGHNE COET/8809/2021 19 34 53 C

88 ADJAKO ONOMROBAKPO SUNNY-OGHENEMINE COET/8213/2021 22 45 67 B

89 AGBATUTU OGHENETEJIRI DANIEL COET/9562/2022 19 29 48 D H.O.D.: Engr. Dr. Enalume K. O.

90 AGBO ANAYO DANIEL COET/9552/2022 10 15 25 F

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN

COLLEGE OF ENGINEERING AND TECHNOLOGY

SECOND SEMESTER

DEPARTMENT: PETROLEUM ENGG ACADEMIC SESSION: 2023/2024

COURSE TITLE: BASIC ELECTRICAL ENGINEERING II COURSE CODE: EEE221

COURSE CREDIT: 2 LEVEL: 200

S/N NAMES MAT NO CA EXAM TOTAL GRADE SUMMARY

91 AGU SUCCESS AMARACHUKWU COET/8389/2021 24 41 65 B GRADE NUMBERS

92 AIYEDUN KENDRICKS OLUWATOBI COET/8266/2021 25 37 62 B A 25

93 AKINDE DAMILOLA OLUWAFEYISAYO COET/9724/2022 22 33 55 C B 50

94 ALBERT DIVINE CHIDALU COET/8615/2021 23 43 66 B C 47

95 AMEH DAVID OJOCHEGBE COET/8270/2021 25 35 60 B D 21

96 AMOS DANIEL ELIJAH COET/9525/2022 22 46 68 B E 48

97 ANAGBE SAMUEL ADEIZA COET/10312/2022 19 24 43 E F 13

98 ANJOLAOLUWA COLLINS OLUWADAMILARE COET/9645/2022 29 51 80 A TOTAL 204

99 ARERE KENNETH OGHENEMARO COET/10030/2022 21 24 45 D

100 ASHIEDU DIVINE ONYEKACHUKWU COET/9502/2022 27 51 78 A PASSED 191

101 AZUBUIKE WALTER EKENE COET/8313/2021 25 40 65 B FAILED 13

102 BALOGUN TOYOSI COET/10340/2022 12 15 27 F

103 BONIFACE MIRACLE CHIDERA COET/8234/2021 28 50 78 A %PASSED 94%

104 BUTUS NEHEMIAH SHEKWOAGA COET/8721/2021 20 30 50 C %FAILED 6%

105 CHARLES HENRY SOMTOCHUKWU COET/8242/2021 21 47 68 B

106 CHIDI MANUCHIMSO FAVOUR COET/9483/2022 19 28 47 D

107 CHIMA VICTOR CHIBUIKE COET/10369/2022 19 35 54 C LECTURER: Engr. Dr. Akinloye B. O.

108 CHUKWU SAMUEL CHRISTIAN COET/9933/2022 21 31 52 C

109 CHUKWUKA EMMANUEL CHIAGOZIELAM COET/10290/2022 9 13 22 F

110 CHUKWUNONYE NOBLE NKEMSINACHUKWU COET/10063/2022 27 44 71 A LECTURER: Engr. Dr. Enalume K. O.

111 EBOHON TOBI PETER COET/10307/2022 22 39 61 B

112 EDILELE EJIRO FAVOUR COET/10392/2022 21 31 52 C

113 EHIEDU LINUS.C. UGOCHUKWU COET/10472/2022 21 33 54 C LECTURER: Engr. Dr. Efenedo

114 EKARO-AKORA SAMUEL AKPEZI COET/10192/202 24 36 60 B

115 EMMANUEL MICHEAL CHINAKASIMOB I COET/10354/2022 21 35 56 C

116 ERHASIE ONOMOESI VICTORY COET/10183/2022 20 33 53 C LECTURER: Engr. Avworo

117 ERUBRENYO FAVOUR COET/8185/2021 10 11 21 F

118 ETUOKWU KIWEN SIMEON COET/10195/2022 25 41 66 B

119 EVANS FAKAOLUWA COET/10247/2022 15 15 30 F H.O.D.: Engr. Dr. Enalume K. O.

120 EWIM PROMISE IKACHUKWU COET/8928/2021 22 40 62 B

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COLLEGE OF ENGINEERING AND TECHNOLOGY

SECOND SEMESTER

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COURSE TITLE: BASIC ELECTRICAL ENGINEERING II COURSE CODE: EEE221

COURSE CREDIT: 2 LEVEL: 200

S/N NAMES MAT NO CA EXAM TOTAL GRADE SUMMARY

121 EZE FRANKLIN CHUKWUEBUKA COET/8271/2021 21 47 68 B GRADE NUMBERS

122 EZEH MATTHEW CHUKWUEMEKA COET/10393/2022 18 29 47 D A 25

123 EZEKIEL GODSON OPOBO COET/9044/2021 21 19 40 E B 50

124 FAMOUS LIBERTY OGHENECHOJA COET/9491/2022 23 17 40 E C 47

125 GAD CHINONSO PRAISE COET/10268/2022 23 39 62 B D 21

126 GEORGE DEBORAH E COET/9592/2022 18 28 46 D E 48

127 GERMAN ISAIAH COET/10188/2022 25 38 63 B F 13

128 GOLDEN OZORO COET/8818/2021 22 36 58 C TOTAL 204

129 IDRIS KAZEEM SHAIBU COET/9444/2022 19 22 41 E

130 IFALOLA GODWIN IYANUOLUWA COET/9413/2022 18 27 45 D PASSED 191

131 IGHARO ETINOSA COET/9245/2021 22 38 60 B FAILED 13

132 IHEME JOSEPH FAVOUR COET/9943/2022 21 29 50 C

133 IJJI WILLIAMS AKOREDE COET/8445/2021 21 47 68 B %PASSED 94%

134 ILESANMI GODSPOWER OLUWADARASIMI COET/10296/2022 21 32 53 C %FAILED 6%

135 ISRAEL ODUVWU PRINCESS ELOHU COET/8967/2021 18 24 42 E

136 IYAJI EMMANUEL ODEY COET/9832/ 2022 22 18 40 E

137 JOHN IFESINACHI CHINEDU COET/9201/2021 22 33 55 C LECTURER: Engr. Dr. Akinloye B. O.

138 KAYODE SAMSON MODUPE COET/10006/2022 25 40 65 B

139 KOLAWOLE KEHINDE EMMANUEL COET/8269/2021 17 24 41 E

140 MADUAGWU DAVID DILICHUKWU COET/8715/2021 21 31 52 C LECTURER: Engr. Dr. Enalume K. O.

141 MADUKA DIVINE OLUOMACHUKWU COET/10287/2022 26 46 72 A

142 MATTHEW TITILAYO MOSEKA COET/8358/2021 23 39 62 B

143 MAXWELL VICTORY CHINELO COET/10038/2022 22 30 52 C LECTURER: Engr. Dr. Efenedo

144 MICHAEL BLESSED GORDIAN COET/10344/2022 18 32 50 C

145 MINIMA SAMSON TUOTAMUNO COET/8300/2021 24 16 40 E

146 MUSA HAFEEZ OYAKHILOME COET/8377/2021 18 22 40 E LECTURER: Engr. Avworo

147 NGWU UDOCHUKWU DAVIS COET/8673/2021 20 32 52 C

148 NIMIBOFAWEI NANAWEBOMOILA ISAAC COET/8496/2021 24 41 65 B

149 NNAMDI CHUKWUEMEKA MATTHEW COET/8731/2021 20 31 51 C H.O.D.: Engr. Dr. Enalume K. O.

150 NWAKAEGHO EMMANUEL ONYEKACHUKWU COET/8352/2021 28 50 78 A

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN

COLLEGE OF ENGINEERING AND TECHNOLOGY

SECOND SEMESTER

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COURSE TITLE: BASIC ELECTRICAL ENGINEERING II COURSE CODE: EEE221

COURSE CREDIT: 2 LEVEL: 200

S/N NAMES MAT NO CA EXAM TOTAL GRADE SUMMARY

151 NWOKO KINGSLEY CHIWENDU COET/10254/2022 18 25 43 E GRADE NUMBERS

152 OBOH FAVOUR OGHENEVWEGBA COET/10267/2022 19 32 51 C A 25

153 ODEH VICTOR OCHE COET/8692/2021 21 32 53 C B 50

154 OFUZOR HENRY IFECHUKWUDENI COET/9664/2022 18 24 42 E C 47

155 OGAR FAVOUR OWOKONU COET/9524/2022 23 32 55 C D 21

156 OGBE GIFT BESTDONE COET/10442/2022 20 20 40 E E 48

157 OGBE OMATSEYE ABLE COET/10498/2022 11 15 26 F F 13

158 OGEDEGBE OKIOGHENE FAITH COET/9492/2022 21 34 55 C TOTAL 204

159 OKETOPE ABDUL QUADRI AYOMIDE COET/10286/2022 19 32 51 C

160 OKOLI CHIDIMMA PRISCA COET/10305/2022 23 30 53 C PASSED 191

161 OKORO PRINCE EZE COET/10360/2022 19 26 45 D FAILED 13

162 OKPIAIFO THERESA OSETUA COET/10323/2022 17 23 40 E

163 OKUBIO UBAROMERA DAVID COET/10299/2022 23 37 60 B %PASSED 94%

164 OLANIYA MICHAEL SEGUN COET/10033/2022 22 32 54 C %FAILED 6%

165 OLANREWaju KAOSARAT PELUMI COET/10056/2022 29 57 86 A

166 OLUWATOBILOBA OLUWATOROMO DAVIDSON COET/8639/2021 26 34 60 B

167 ONYEBBULA ROSELINE UCHECHI COET/9478/2022 27 47 74 A LECTURER: Engr. Dr. Akinloye B. O.

168 ONYEKWERE DAVID FAVOUR COET/9421/2022 18 28 46 D

169 ONYIA FAVOUR COET/8543/2021 25 43 68 B

170 OPONE CHUKWUDI JAPHETH COET/8874/2021 29 47 76 A LECTURER: Engr. Dr. Enalume K. O.

171 OSOGWEOVO PRAISE OJEGBA COET/8811/2021 18 23 41 E

172 OWOEYE TOBILOBA SAMUEL COET/10329/2022 8 12 20 F

173 OZOCHI CHIOMA MARYROSE COET/10266/2022 29 56 85 A LECTURER: Engr. Dr. Efenedo

174 PAUL EMOGHENE COET/8288/2021 28 51 79 A

175 SALAU VICTOR SAMSON COET/8243/2021 19 22 41 E

176 SAMUEL OKOOBOH COET/10285/2022 20 34 54 C LECTURER: Engr. Avworu

177 UCHENDU CALEB CHINEMEREMU COET/10405/2022 27 45 72 A

178 UDEGBOUHAN CHINECHINDU MICHEAL COET/10364/2022 16 24 40 E

179 UKWUMONU PRINCE COET/8559/2021 24 17 41 E H.O.D.: Engr. Dr. Enalume K. O.

180 VICTOR SAMUEL AZIBALUA COET/10317/2022 21 29 50 C

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COLLEGE OF ENGINEERING AND TECHNOLOGY

SECOND SEMESTER

DEPARTMENT: PETROLEUM ENGG ACADEMIC SESSION: 2023/2024

COURSE TITLE: BASIC ELECTRICAL ENGINEERING II COURSE CODE: EEE221

COURSE CREDIT: 2 LEVEL: 200

S/N NAMES MAT NO CA EXAM TOTAL GRADE SUMMARY

181 VITALIS BLESSING OSINACHI COET/8328/2021 20 25 45 D GRADE NUMBERS

182 YUSUF YUSUF AKEEM COET/8214/2021 17 23 40 E A 25
183 NDEGO FRANK COET/8488/2021 19 26 45 D B 50
184 SAMUEL OJOSHIMITE STELLA COET/8838/2021 24 44 68 B C 47
185 OGIORIO NELSON IKPONMWOSA COET/10150/2022 19 33 52 C D 21
186 CHIKEZIE VICTOR KELECHI COET/7293/2020 21 33 54 C E 48
187 OBI MICHAEL KENWA COET/7334/2020 25 43 68 B F 13
188 ONOME GODSWILL OREVAOGHENE COT/5200/2018 20 35 55 C TOTAL 204
189 NWACHUKWU SUNDAY ERIC COET/7327/2020 18 29 47 D
190 OFUOMADEGA DIFFERENCE COET/7338/2020 20 28 48 D PASSED 191
191 AKPOVEGNWARE EFEMENA ALFRED COET/7282/2020 24 39 63 B FAILED 13
192 ESEVWERHERE OGHENETEGA PRAISE COT/5324/2018 6 7 13 F
193 IGBEN OGHENEVWORO GEOFFREY COET/7314/2020 23 17 40 E %PASSED 94%
194 GUWOR PREYE COT/2507/2016 22 18 40 E %FAILED 6%
195 OKATENWU OHINOYI ABULLAHI COT/5315/2018 21 36 57 C
196 ISHIOKU FAVOUR COT/5274/2018 22 36 58 C
197 TIOKPASSON ELIJAH O. COT/5263/2018 16 25 41 E LECTURER: Engr. Dr. Akinloye B. O.
198 ALADETAN ADEKANMI JEREMIAH COT/3921/2017 19 22 41 E
199 VICTOR CHUKWUDEBE ONYEKA COT/2468/2016 19 34 53 C
200 UGOH TITUS HENENTER COET/7366/2020 19 28 47 D LECTURER: Engr. Dr. Enalume K. O.
201 MICHAEL ELVIS COT/5265/2018 18 24 42 E
202 MBANEFO ZENO CHIMUANYA COET/10488/2022 19 29 48 D
203 PRINCEWILL KELECHI BLESSING COT/5296/2018 22 46 68 B LECTURER: Engr. Dr. Efenedo
204 EZE PROSPER CHIDERA COET/10269/2022 21 35 56 C

LECTURER: Engr. Avworo

H.O.D.: Engr. Dr. Enalume K. O.

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Paper 6 of 6 — EEE 221 BASIC ELECTRICAL ENGINEERING II (2022/2023)

Level 200 • Second Semester • Handout

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date

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF TECHNOLOGY
EXAMINATION: First Semester
COURSE CODE: EEE 211
TIME ALLOWED: 2 Hrs



SESSION: 2018/2019
COURSE TITLE: Basic Electrical Engineering I
DEGREE: B.ENG

INSTRUCTION: Answer FOUR Questions from both sections

SECTION A: Answer TWO Questions from this section

1. (a) State Maximum Power Transfer theorem, and justify the need for such a theorem. (4½ marks)
- (b) (i) State Superposition theorem. Explain why this theorem does not apply to power as an electrical quantity, and state how power can be computed using this theorem. (5 marks)
- (ii) Using Superposition theorem, calculate the current flowing through the $4\ \Omega$ resistor in figure Q1 (b). Also, compute the voltage drop and power dissipated in the $4\ \Omega$ resistor. (8 marks)

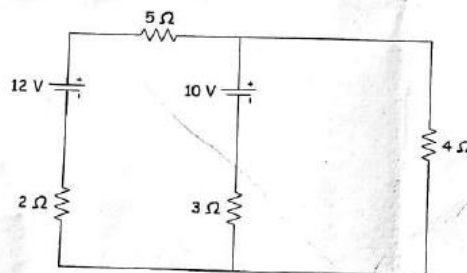


Figure Q1 (b)

2. (a) Explain the following terms: (i) Node (ii) Loop (iii) Branch (iv) Transient response. (6 marks)
- (b) For the circuit of figure Q2 (b), (i) Write out the differential equation (ii) Compute the time constant, natural response and forced response (3½ marks)
- (c) Calculate the mesh currents and branch currents in the circuit of figure Q2 (c). (8 marks)

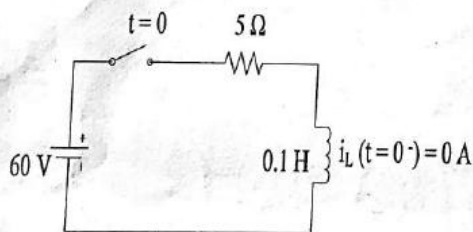


Figure Q2 (b)

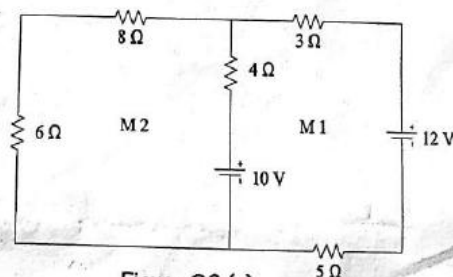


Figure Q2 (c)

3. (a) State Kirchhoff's current law and Kirchhoff's voltage law. (3 marks)
- (b) State Norton's theorem. (2½ marks)
- (c) (i) Using Thevenin theorem, calculate the current through the $8\ \Omega$ resistor in figure Q3 (c). (9 marks)
- (ii) Draw the Norton equivalent circuit of the circuit obtained in Question 3c (i) and verify the value of the current flowing through the $8\ \Omega$ resistor. (3 marks)

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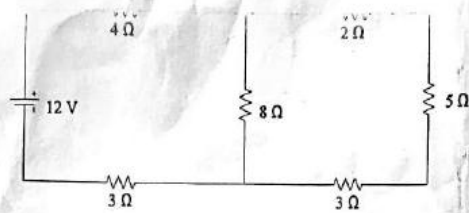


Figure Q3 (c)

SECTION B: Answer TWO Questions from this section

4. (a) State the laws of magnetism. (3½ marks)
- (b) Define the following: (i) magnetic field (ii) magnetic flux (iii) relative permeability (6 marks)
- (c) An inductive coil of resistance $10\ \Omega$ and inductance 0.1 H is connected in parallel with a $150\ \mu\text{F}$ capacitor to a variable frequency, 200 V supply. Find the resonant frequency at which the total current taken from the supply is in phase with the supply voltage. Find also the value of this current. Draw the circuit arrangement and the phasor diagrams. (8 marks)
5. (a) With the aid of a circuit diagram and all necessary equations, show that when an alternating voltage is applied to a purely capacitive circuit, the resulting current leads the applied voltage by $\frac{\pi}{2}$ radians. (8 marks)
- (b) A magnetic circuit is excited by three coils as shown in figure Q5 (b). Calculate the flux produced in the air gap. The material used for core is iron having relative permeability of 800. The length of the magnetic circuit is 100 cm with an air gap of 2 mm in it. The core has uniform cross-section of 6 cm^2 . Take $\mu_0 = 4\pi \times 10^{-7}$. (9½ marks)

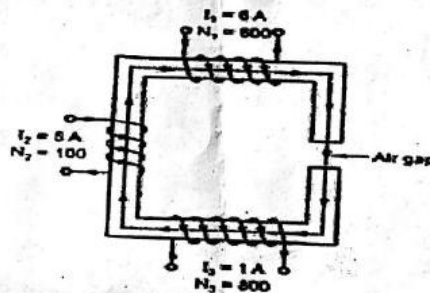


Figure Q5 (b)

6. (a) Enumerate the importance of RMS value in the field of Engineering. (6 marks)
- (b) The waveform of the voltage and current of a circuit are given by:
 $e = 120\sin(314t)$ and $i = 10\sin(314t + \frac{\pi}{6})$. Calculate the values of the resistance and capacitance which are connected in series to form the circuit. Also draw waveforms for current, voltage and phasor diagram. Calculate power consumed by the circuit. (11½ marks)

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